

COMPARISON OF TRADITIONAL ML AND AI

1. Create a table comparing traditional AI/ML and Gen AI across at least five aspects such as data requirements, output type, human interaction, adaptability, and example use cases. Use real-world apps you know (like Instagram, Zomato, or Spotify) as examples for each.

Ans. Aspect: Data Requirements

Traditional AI/ML: Uses labeled data for training and prediction.

Generative AI: Uses large datasets to learn patterns and generate new content.

Aspect: Output Type

Traditional AI/ML: Produces predictions, classifications, or recommendations.

Generative AI: Produces new text, images, audio, videos, or code.

Aspect: Human Interaction

Traditional AI/ML: Limited interaction and usually follows predefined rules.

Generative AI: Supports natural and dynamic conversations with users.

Aspect: Adaptability

Traditional AI/ML: Works well for specific tasks but has limited flexibility.

Generative AI: Can adapt to different tasks and generate diverse outputs.

Aspect: Example Use Case

Traditional AI/ML: Spotify song recommendations based on listening history.

Generative AI: ChatGPT generating personalized playlist descriptions.

Aspect: Real-World App Example

Traditional AI/ML: Zomato restaurant recommendation system.

Generative AI: AI chatbot generating restaurant suggestions and explanations.

2. Given a scenario where Zomato recommends restaurants based on your previous orders, explain whether this uses traditional AI/ML or Gen AI, and justify your answer in 3-4 sentences.

Ans. When Zomato recommends restaurants based on my previous orders, it mainly uses Traditional AI/ML. The system analyzes my order history, ratings, location, and preferences to predict which restaurants I may like. Its goal is to make recommendations rather than create new content. Therefore, this feature is primarily a machine learning recommendation system, although GenAI could be added later to explain recommendations in a conversational way.

3. Pick any feature from an app you use daily (such as Instagram's photo filters or Spotify's song recommendations) and describe how it could be improved or transformed using Gen AI instead of traditional AI/ML. Write 4-5 lines explaining the change.

Ans. Feature Chosen: Spotify Song Recommendations

Currently, Spotify recommends songs based on listening history and user behavior using traditional machine learning techniques. With Generative AI, Spotify could create personalized playlists for specific moods, events, or activities and generate descriptions explaining why each song was selected. It could also create custom introductions for playlists and suggest new music in a conversational manner. This would make the music discovery experience more interactive and personalized for users.

4. Imagine you are building a chatbot for Flipkart customer support. List two ways a traditional AI/ML approach would handle

user queries, and two ways a Gen AI approach would differ in handling the same queries. Hint: Focus on how responses are generated and the flexibility of the answers.

Ans. Traditional AI/ML Approach

1. The chatbot would match user queries with predefined keywords and provide fixed responses from a database.
2. It would classify queries into categories such as order tracking, returns, or payments and display standard answers.

Generative AI Approach

1. The chatbot would generate natural, human-like responses tailored to each customer's specific question.
2. It could understand complex queries, maintain conversation context, and provide detailed explanations or personalized solutions instead of fixed responses.

Key Difference

Traditional AI mainly selects answers from predefined options, while Generative AI creates new responses dynamically based on the user's request and conversation context.